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Review

Attention-enhanced reinforcement learning for dynamic portfolio optimization

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ABSTRACT

We propose a deep reinforcement learning framework for dynamic portfolio optimization that combines a Dirichlet policy with cross-sectional attention mechanisms. The Dirichlet distribution enforces feasibility by construction, accommodates tradability masks, and provides a coherent geometry for exploration. Our architecture integrates per-asset temporal encoders with a global attention layer, allowing the policy to adaptively weight sectoral co-movements, factor spillovers, and other cross-asset dependencies. We evaluate the framework on a comprehensive S&P 500 panel from 2000 to 2025 using purged walk-forward backtesting to prevent look-ahead bias. Empirical results show that attention-enhanced Dirichlet policies deliver higher terminal wealth, Sharpe and Sortino ratios than equal-weight and reinforcement learning baselines, while maintaining realistic turnover and drawdown profiles. Our findings highlight that principled action parameterization and attention-based representation learning materially improve both the stability and interpretability of reinforcement learning methods for portfolio allocation.

1. Introduction

Designing robust portfolio allocation strategies under uncertainty is a central problem in financial economics. Classical frameworks, beginning with mean–variance optimization (Markowitz, 1952) and intertemporal extensions (Merton, 1971), prescribe allocations by trading off expected return against risk. While these models are analytically elegant, they rely on restrictive assumptions about return distributions, stationarity, and investor preferences, and they typically neglect frictions such as transaction costs. Subsequent work has refined the measurement of risk using tail-sensitive criteria (e.g., CVaR, drawdown) and improved the estimation of covariance matrices through shrinkage, factor structures, and robust optimization. Yet, even with these advances, most approaches remain essentially static or myopic, treating allocation as a one-period problem solved repeatedly rather than as a dynamic, sequential decision under evolving market conditions.

Reinforcement learning (RL) offers a natural framework for casting portfolio choice as a Markov decision process (MDP). In this setting, the investor observes a state summarizing recent market history, chooses an action corresponding to portfolio weights, and receives a reward defined as portfolio performance net of frictions. Policy-gradient and actor–critic algorithms can then be used to directly optimize long-run performance. A growing literature has applied RL to trading and portfolio management, with encouraging empirical results (Jiang et al., 2017; Moody & Saffell, 2001; Yang et al., 2020; Zhang et al., 2020). However, existing studies often rely on ad-hoc action parameterizations (e.g., mapping logits through a softmax and projecting onto the simplex), omit realistic trading frictions, or evaluate performance in limited settings.

In this paper we revisit the portfolio RL problem with a focus on principled action parameterization and rigorous economic evaluation. We propose a flat Dirichlet policy that parameterizes the distribution of portfolio weights directly on the simplex. This design enforces the budget and non-negativity constraints by construction, handles tradability masks naturally, and provides a coherent geometry for exploration in policy gradients. We embed economically motivated frictions into the reward: per-period turnover costs to capture transaction costs, and a quadratic variance penalty $w_t^\top \Sigma_t w_t$ to regularize concentration in correlated assets. This connects the RL objective to classical portfolio trade-offs while retaining sequential adaptivity. We implement a rigorous evaluation protocol: walk-forward backtesting with purging to prevent look-ahead bias, reporting of after-cost performance metrics (Sharpe, Sortino, drawdown, turnover), and factor regressions to assess whether learned strategies capture genuine alpha rather than rediscovering known exposures.

By combining a principled policy distribution, realistic cost and risk modeling, and stringent evaluation, our study clarifies what reinforcement learning contributes to portfolio management when stripped

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of ad-hoc design choices. While our focus in this first paper is on a *non-hierarchical* (flat) action structure to isolate these ingredients, the framework is extensible: richer state encoders (e.g., temporal transformers, cross-asset attention) and hierarchical action decompositions can be layered atop this foundation. Our results show that even in the flat setting, enforcing feasibility at the action level and embedding economic frictions materially improve stability and economic realism relative to standard RL baselines.

2. Related literature

2.1. Foundations of portfolio choice and risk management

The modern theory of portfolio choice begins with mean–variance optimization (Markowitz, 1952), later extended to intertemporal settings (Merton, 1971) and growth-optimal strategies that emphasize long-run geometric growth (Kelly, 1956). Building on these foundations, the literature addresses portfolio risk through both measurement and regularization. On the measurement side, while variance remains the workhorse criterion, tail-sensitive measures like Conditional Value-at-Risk (CVaR) (Rockafellar & Uryasev, 2000) and path-dependent drawdown objectives (Grossman & Zhou, 1993) are increasingly used to capture asymmetric downside risk. On the regularization side, estimating risk in high dimensions requires shrinkage estimators (Ledoit & Wolf, 2004) or factor-based structures to mitigate noise in sample covariance matrices. Furthermore, dynamic models emphasize that transaction costs and implementation shortfall materially alter optimal policies (Almgren & Chriss, 2000; Gatheral, 2010), necessitating rebalancing schedules that internalize frictions. Our reward function connects to these traditions by integrating log-utility with quadratic variance penalties and turnover-based transaction costs.

2.2. Cross-sectional predictability and implementation constraints

Expected returns exhibit persistent cross-sectional patterns across factors such as value, momentum, and volatility clustering (Bollerslev, 1986; Fama & French, 1993; Jegadeesh & Titman, 1993), though the proliferation of signals raises significant data-mining concerns (Harvey et al., 2016). Contemporary portfolio construction combines these signals while managing risk and implementation frictions through constrained optimization. A related line of research utilizes metaheuristic methods, including evolutionary search and swarm intelligence, to solve index-tracking problems subject to cardinality and turnover limits (Silva & Almeida Filho, 2023; Yuen et al., 2021, 2022). These studies highlight the importance of constraint-aware design, a principle we incorporate through our use of tradability masks and turnover-sensitive reward shaping.

2.3. From control to learning: MDPs and RL for portfolio management

Casting portfolio choice as a Markov Decision Process (MDP) allows one to optimize sequentially under uncertainty while accounting for state evolution, risk, and costs. Early work investigated recurrent reinforcement learning for trading (Moody & Saffell, 2001) and order execution (Nevmyvaka et al., 2006). With deep neural networks, a wave of studies explored policy gradient and actor–critic methods for asset allocation and trading (e.g., Deng et al., 2016; Jiang et al., 2017; Yang et al., 2020; Zhang et al., 2020). The appeal of RL lies in optimizing a directly specified, path-dependent objective (e.g., log growth net of costs) without requiring fully specified models for returns. Methodologically, our approach aligns with policy-gradient reinforcement learning, particularly proximal policy optimization (PPO) (Schulman et al., 2017), but departs from many prior portfolio RL studies in two key ways: (i) we model the action as a probability distribution on the simplex via a Dirichlet policy, ensuring feasibility and providing a principled exploration geometry; and (ii) we explicitly incorporate transaction costs and a covariance-based risk penalty in the per-period reward, tying the learning objective to standard economic trade-offs.

2.4. Action parameterizations for simplex-constrained decisions

Most RL implementations for portfolio weights map unconstrained network outputs through a softmax to the simplex, sometimes followed by projection to enforce masks. This approach couples exploration across all coordinates and may induce unstable gradients when post-hoc projections are used. In contrast, Dirichlet policies parameterize a distribution supported exactly on the simplex, with concentration parameters controlling dispersion and hence exploration in a geometrically appropriate way. Dirichlet parameterizations have been studied for constrained action spaces in continuous-control RL and Bayesian decision-making; they provide analytic log-probabilities and reparameterized sampling for policy-gradient updates.

2.5. Representation learning: Sequence models and attention in finance

Financial time-series are characterized by long memory, regime shifts, and complex cross-sectional co-movements. Deep sequence models such as CNNs and LSTMs have traditionally been used to encode these features, yet attention-based architectures like temporal fusion transformers and long-horizon variants (Lim et al., 2021; Zhou et al., 2021) now provide state-of-the-art performance by adaptively focusing on salient temporal lags and cross-asset dependencies. Our framework leverages this evolution by employing a high-dimensional rolling window tensor $X_{t-W+1:t}$ as the state input. By utilizing a flat Dirichlet policy head, we isolate the impact of this representation learning while maintaining a modular design that can accommodate increasingly sophisticated temporal or graph-based encoders.

2.6. Risk, costs, and constraints in learning-based allocation

A practical portfolio algorithm must internalize transaction costs, capacity limits, and regulatory constraints. Turnover penalties and no-trade bands are common in industry portfolio construction; Almgren–Chriss style impact models inform cost calibration (Almgren & Chriss, 2000). Constraints (no shorting, position caps, leverage caps) are typically enforced by projection or via regularization in the optimizer. In RL, encoding constraints at the action level reduces training pathologies. Our design enforces the budget and non-negativity constraints by construction (Dirichlet support), masks untradable assets directly in the policy, and optionally applies per-name caps via capped-simplex projection. The risk penalty $w^\top \Sigma w$ regularizes concentration in correlated names and connects the RL objective to classical mean–variance preferences.

2.7. Backtesting pitfalls and evaluation protocols

Backtest overfitting and data-snooping are pervasive dangers in financial ML (Bailey et al., 2017, 2016; Harvey et al., 2016). Robust evaluation emphasizes (i) realistic frictions; (ii) walk-forward splits with embargo to prevent leakage; (iii) out-of-sample reporting of performance (Sharpe/Sortino, MDD, turnover) *after* costs; (iv) factor regressions to attribute alpha; and (v) reality-check or SPA tests for multiple hypothesis correction (Romano & Wolf, 2005; White, 2000). We adopt a purged walk-forward design with transaction costs built into the reward and report factor-adjusted performance to assess whether the learned policy captures genuine alpha rather than rediscovering standard factor exposures.

Table 1
Summary of the S&P 500 daily panel dataset.

Feature	Value	Notes
Time span	01/04/2000–08/20/2025	Daily adjusted data
Trading days (T)/Tickers (N)	6446/482	Historical constituents
Avg. tradable names per day	389.5	Accounting for entry/exit
State window (W)/Features (F)	30/19	Rolling window inputs
Transaction cost (κ)	5 bps	Per dollar turnover

2.8. Positioning of our contribution

To summarize, our work sits at the intersection of: (a) classical dynamic portfolio choice with risk and costs; (b) modern RL for sequential allocation; and (c) representation learning for multivariate financial time-series. Relative to prior portfolio RL studies that use softmax heads or projection for feasibility, we contribute a *flat Dirichlet policy* that (i) respects the simplex and masks by construction; (ii) provides a coherent exploration geometry for policy gradients; and (iii) integrates directly with an economically motivated reward (log returns net of transaction costs and variance penalty). The non-hierarchical focus in this first paper isolates these ingredients and clarifies their impact under rigorous, out-of-sample evaluation.

3. Data and preprocessing

3.1. Universe and feature engineering

We utilize a daily panel of U.S. equities from the S&P 500 universe spanning 2000–2025. To mitigate survivorship bias, the sample includes historical index constituents rather than a static recent list, with adjusted prices to account for corporate actions. The raw data is transformed from a long-format CSV into a three-dimensional tensor $X \in \mathbb{R}^{T\times N\times F}$, where $T = 6446$ trading days, $N = 482$ unique tickers, and $F = 19$ features including volume and technical indicators. Percentage returns $r_{t,i}$ are derived from adjusted close prices via log-returns $\mathrm{LR}_{t,i} = \log(\mathrm{Close}_{t,i}/\mathrm{Close}_{t-1,i})$. To stabilize learning while preserving cross-sectional signals, we apply daily z-scoring within each feature:

$$Z_{t,i,f} = \frac{X_{t,i,f} - \mu_{t,f}}{\sigma_{t,f} + 10^{-8}}, \tag{1}$$

where $\mu_{t,f}$ and $\sigma_{t,f}$ are the cross-sectional mean and standard deviation of feature f across all tradable assets at time t.

3.2. Environmental state and evaluation protocol

The reinforcement learning state is a rolling window $s_t = \{Z_{t-W+1:t}, M_t, m_t\}$, where $W = 30$ days, $M_t \in \{0,1\}^N$ is a binary tradability mask ensuring allocations only to available assets, and m_t denotes optional market-level covariates. Table 1 summarizes the dataset parameters. To ensure rigorous evaluation, we employ a purged walk-forward protocol with a chronological split: 2000–2016 for training, 2017–2019 for validation, and 2020–2025 for testing.

This temporal structure is isolated via a $W + 1$ day purge length and a 30-day embargo following each block to prevent look-ahead bias and data leakage. The agent rebalances daily at market close, with risk penalties calculated using a rolling $L = 60$ day covariance window Σ_t.

4. Economic formulation and environment

4.1. MDP formulation and state space

We cast the daily portfolio allocation problem as a finite-horizon Markov Decision Process (MDP) where an agent seeks to maximize the expected discounted sum of rewards. The state s_t at each trading day t is represented as a structured object:

$$s_t = \{X_{t-W+1:t}, M_t, m_t\} \tag{2}$$

where $X_{t-W+1:t} \in \mathbb{R}^{W\times N\times F}$ is a tensor capturing the last W days of observations across N assets and F features. This rolling window allows the agent to model temporal dependencies such as momentum and volatility clustering. The state further includes a tradability mask $M_t \in \{0,1\}^N$ to handle delistings or suspensions and market-level covariates $m_t \in \mathbb{R}^K$ to capture systemic conditions. By conditioning on this rich information set, we approximate the Markov property, assuming that the distribution of next-period returns is independent of history prior to the window W.

4.2. Action space and Dirichlet parameterization

The action a_t corresponds to the choice of a portfolio vector $w_t \in \Delta^{N+1}$ on the unit simplex, where $w_{t,0}$ denotes the cash allocation and $w_{t,1:N}$ represents risky asset weights. Unlike standard approaches that map unconstrained logits through a softmax, we model actions as draws from a flat Dirichlet distribution:

$$w_t \sim \mathrm{Dirichlet}(\alpha_0, \alpha_1, \dots, \alpha_N) \tag{3}$$

This parameterization enforces budget and non-negativity constraints by construction, naturally accommodates tradability masks, and provides a principled geometry for exploration via the concentration parameters α_i.

4.3. Reward function and transition dynamics

The reward R_t connects the reinforcement learning objective to classical mean–variance trade-offs by penalizing transaction costs and risk:

$$R_t = \log(1 + w_{t,0} r_t^{\mathrm{cash}} + w_{t,1:N}^\top r_t) - \kappa \sum_{i=1}^{N} |w_{t,i} - \tilde{w}_{t,i}^{\mathrm{pre}}| - \lambda w_t^\top \Sigma_t w_t \tag{4}$$

The first term captures compounding portfolio growth, where r_t^{cash} represents the time-varying risk-free rate. The second term penalizes turnover at a rate $\kappa > 0$ per dollar traded, while the final term applies a quadratic penalty based on the rolling covariance matrix Σ_t to discourage excessive concentration in correlated assets.

Transitions occur as the agent selects weights w_t, asset returns r_t are realized, and the state window rolls forward to $t+1$. This structure accounts for price-driven weight drift and ensures the agent internalizes the impact of current rebalancing on future trading costs.

5. Method and policy architecture

5.1. Overview

This section specifies a stochastic policy for daily portfolio choice that maps observed market state to a *distribution* over portfolio weights on the simplex. The architecture has three blocks: (i) a *per-asset temporal encoder* that turns a $W\times F$ window into an embedding, (ii) a *cross-sectional mixer* that models relations across assets at date t, and (iii) a *single Dirichlet action head* over [cash + all assets].

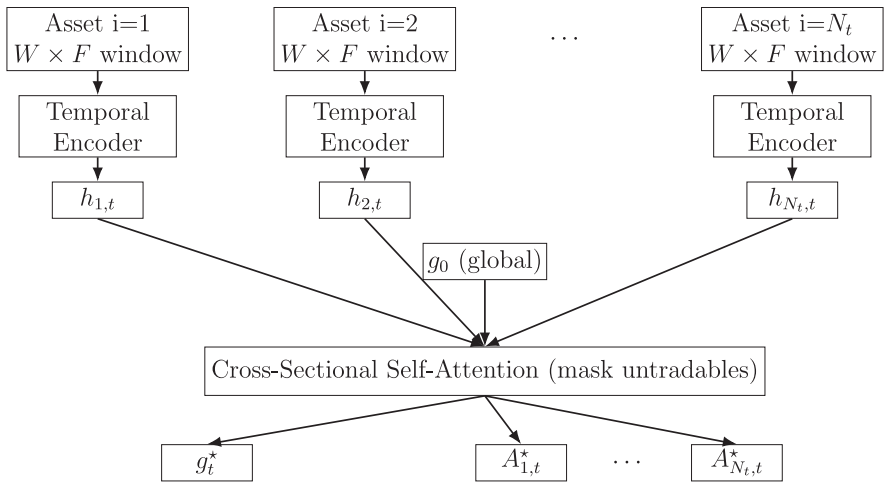

Fig. 1. Temporal→token and cross-sectional attention encoder.

5.2. Setting and notation

At date t there are N_t tradeable equities. For each asset i we observe a $W \times F$ window of features

$$X_{i,t-W+1:t} \;=\; [\, x_{i,t-W+1}, \dots, x_{i,t} \,] \in \mathbb{R}^{W \times F},$$

a tradability mask $m_t \in \{0,1\}^{N_t}$ (1 if tradeable), and optional market factors $z_t \in \mathbb{R}^K$. The state is

$$s_t \;=\; \big(X_{1,t-W+1:t}, \dots, X_{N_t,t-W+1:t},\; m_t,\; z_t\big).$$

Actions are portfolio weights $w_t \in \mathbb{R}_{\geq 0}^{N_t+1}$ (index 0 denotes cash), subject to

$$\sum_{i=0}^{N_t} w_{t,i} = 1, \qquad w_{t,i} = 0 \;\text{ if }\; m_{t,i} = 0 \;\; (i \geq 1).$$

The policy is a conditional distribution $\pi_\theta(\cdot \mid s_t)$ over such w_t.

5.3. Encoders: Temporal (time → token) and cross-sectional (assets ↔ assets)

For each asset i, features $x_{i,\tau} \in \mathbb{R}^F$ over a rolling window $\tau = t - W + 1, \dots, t$ are first projected to model width d via a shared linear map $W_{\text{in}} \in \mathbb{R}^{F \times d}$,

$$u_{i,\tau} \;=\; x_{i,\tau}\, W_{\text{in}} \in \mathbb{R}^d.$$

A shared temporal Transformer encoder is then applied independently to each asset. We add sinusoidal positional encodings $P \in \mathbb{R}^{W \times d}$ and apply L_{time} self-attention layers along the time dimension,

$$Z_{i,\tau}^{(0)} = u_{i,\tau} + P_\tau, \quad Z_i^{(\ell+1)} = \text{LN}\big(Z_i^{(\ell)} + \text{MHSA}(Z_i^{(\ell)})\big),$$
$$Z_i^{(\ell+1)} = \text{LN}\big(Z_i^{(\ell+1)} + \text{FFN}(Z_i^{(\ell+1)})\big).$$

The asset-level embedding $h_{i,t} \in \mathbb{R}^d$ is obtained by pooling across the W temporal tokens (last or mean). Encoder weights are shared across assets, yielding the cross-sectional token matrix

$$H_t \;=\; [\, h_{1,t}, \dots, h_{N_t,t} \,]^\top \in \mathbb{R}^{N_t \times d}.$$

Fig. 1 summarizes the information flow from per-asset temporal windows to latent embeddings $\{h_{i,t}\}$, the construction of the global token g_0, and the permutation-invariant cross-sectional self-attention mechanism that produces the global summary $g_t^\star$ and asset-level representations $A_t^\star$, highlighting shared temporal encoding, asset masking prior to interaction, and global–local information aggregation.

Cross-sectional mixer. To capture co-movement and relative strength across assets on date t, we apply a Transformer encoder *across* tokens. Prepend a learnable global token $g_0 \in \mathbb{R}^d$ and form

$$T_t^{(0)} \;=\; [\, g_0;\; h_{1,t}; \dots; h_{N_t,t} \,] \in \mathbb{R}^{(1+N_t) \times d}.$$

Apply $L_\times$ encoder layers with self-attention across tokens (no positional encodings on assets to preserve permutation invariance), masking untradable asset tokens:

$$Q = T_t^{(\ell)} W_Q,\; K = T_t^{(\ell)} W_K,\; V = T_t^{(\ell)} W_V,$$
$$A = \text{softmax}\Big(\frac{QK^\top}{\sqrt{d_k}} + M_\times\Big),\; U = AV,$$
$$\tilde{T}_t^{(\ell)} = \text{LN}(T_t^{(\ell)} + UW_O), \quad T_t^{(\ell+1)} = \text{LN}(\tilde{T}_t^{(\ell)} + \text{FFN}(\tilde{T}_t^{(\ell)})).$$

Let $T_t^\star = T_t^{(L_\times)}$ and extract the *global summary* $g_t^\star = T_t^\star[0] \in \mathbb{R}^d$ and *asset summaries* $A_t^\star = T_t^\star[1:] \in \mathbb{R}^{N_t \times d}$. Optionally inject macro factors via $g_t^\star \leftarrow g_t^\star + W_{\text{mkt}} z_t$ with $W_{\text{mkt}} \in \mathbb{R}^{K \times d}$.

Interpretability of attention. Cross-sectional attention provides a transparent measure of how the model allocates informational focus across assets when forming portfolio representations. Each attention coefficient reflects the relative influence of one asset's return and feature dynamics on another's representation. In periods of high cross-sectional dispersion, attention concentrates on assets or sectors that explain a larger share of contemporaneous variation, highlighting dominant drivers of portfolio risk and expected return; when dispersion is low or correlations are high, attention weights become more uniform, indicating diffuse information. This adaptive reweighting offers an economically interpretable view of shifting factor exposures and cross-asset dependencies, analogous to how active portfolio managers reallocate analytical emphasis as market conditions evolve.

5.4. Action head: Single Dirichlet over [cash + all assets]

From $(g_t^\star, A_t^\star)$ the actor produces logits for cash and each asset:

$$\ell_t^{(\text{cash})} = a_{\text{cash}}^\top g_t^\star + b_{\text{cash}} \in \mathbb{R},$$
$$\ell_{t,i}^{(\text{asset})} = a_{\text{asset}}^\top A_{t,i}^\star + b_{\text{asset}} \in \mathbb{R},\; i = 1, \dots, N_t.$$

Stack and map to Dirichlet concentrations (with small $\varepsilon > 0$ for stability):

$$\alpha_t^{(\text{flat})} \;=\; \text{softplus}\Big(\, [\, \ell_t^{(\text{cash})},\; \ell_{t,1}^{(\text{asset})}, \dots, \ell_{t,N_t}^{(\text{asset})} \,] \,\Big) + \varepsilon \;\in\; \mathbb{R}_{>0}^{N_t+1}.$$

Sample during training (or use the mean at evaluation):

$$p_t^{(\text{flat})} \sim \text{Dirichlet}\big(\alpha_t^{(\text{flat})}\big), \qquad \mathbb{E}\big[p_t^{(\text{flat})}\big] = \alpha_t^{(\text{flat})} / (\mathbf{1}^\top \alpha_t^{(\text{flat})}).$$

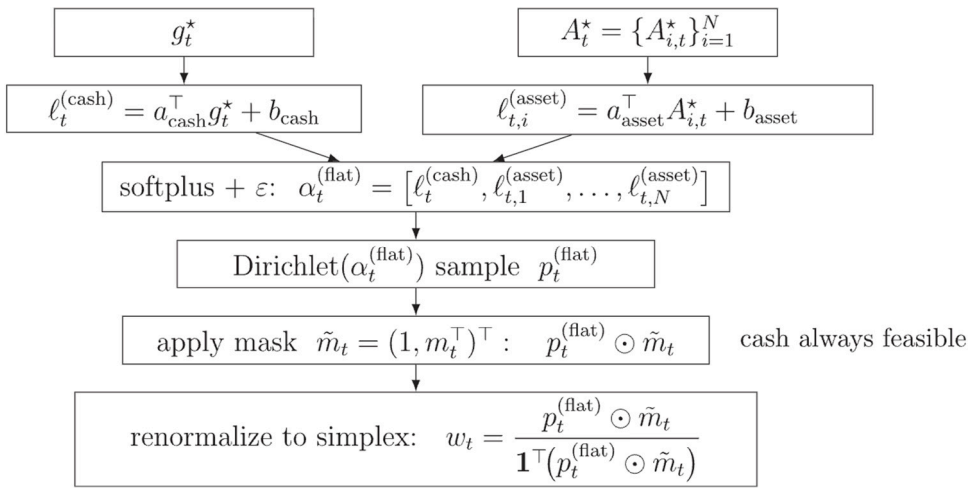

Fig. 2. Dirichlet action head.

Apply the tradability mask ($\tilde{m}_t = (1, m_t^\top)^\top$; cash always feasible) and renormalize:

$$w_t = \frac{p_t^{(\text{flat})} \odot \tilde{m}_t}{\mathbf{1}^\top\left(p_t^{(\text{flat})} \odot \tilde{m}_t\right)} \in \Delta^{N_t}.$$

To summarize the computation graph of the Dirichlet action head, Fig. 2 illustrates how the per-token logits are transformed into concentration parameters, how the Dirichlet sample (or mean) is mapped to feasible portfolio weights through masking and renormalization, and how cash feasibility is handled explicitly. This schematic provides a visual counterpart to the algebraic description above and highlights the role of the Dirichlet parameterization in ensuring simplex feasibility and principled exploration.

5.5. Critic and PPO training

A value head $V_\phi(s_t)$ is computed from $g_t^\star$. With advantages $\hat{A}_t$ (GAE-λ) and old log-probs $\log \pi_{\theta_{\text{old}}}(a_t \mid s_t)$, PPO maximizes

$$\max_\theta \ \mathbb{E}\Big[\min\Big(\rho_t(\theta)\,\hat{A}_t,\ \text{clip}(\rho_t(\theta), 1-\epsilon, 1+\epsilon)\,\hat{A}_t\Big)\Big] - \eta\,\text{KL}\big(\pi_\theta \| \pi_{\theta_{\text{old}}}\big),$$
$$\text{where} \quad \rho_t(\theta) = \exp\big(\log \pi_\theta(a_t|s_t) - \log \pi_{\theta_{\text{old}}}(a_t|s_t)\big). \tag{5}$$

with value loss $\frac{1}{2}(V_\phi(s_t) - \hat{V}_t)^2$ and optional Dirichlet entropy. The policy log-density is

$$\log \pi_\theta(a_t \mid s_t) = \log \text{Dirichlet}\big(p_t^{(\text{flat})};\ \alpha_t^{(\text{flat})}\big),$$

evaluated at the (sampled) pre-mask $p_t^{(\text{flat})}$ used to form w_t.

5.6. Masking, constraints, and frictions

Budget and non-negativity hold by construction; tradability is enforced via $\tilde{m}_t$ and renormalization. Optional name caps are enforced by projecting w_t onto a capped simplex. Transaction costs enter the per-period reward through turnover; optional risk penalties use a rolling covariance estimated from recent returns.

5.7. Complexity and engineering notes

Temporal Transformer as $\mathcal{O}(B\,N_t(W\,d^2 + W^2 d))$. Cross-sectional attention scales as $\mathcal{O}(B\,(N_t{+}1)^2 d)$. We use (i) a learned input projection W_{in} so F can change, (ii) asset-wise chunking in the temporal encoder, and (iii) masking to keep memory bounded when N_t is large.

5.8. Flat vs. hierarchical Dirichlet policies

Formally, we employ a *flat* Dirichlet policy that models the entire portfolio as a single distribution over all tradable assets and cash:

$$w_t \sim \text{Dirichlet}\big(\alpha_t^{(\text{flat})}\big), \quad \alpha_t^{(\text{flat})} \in \mathbb{R}_{>0}^{N_t+1}, \qquad \sum_{i=0}^{N_t} w_{t,i} = 1, \ \ w_{t,i} \geq 0.$$

This construction enforces feasibility on the simplex by design and allows the policy network to learn cross-asset dependencies directly through attention, while remaining parameter-efficient with straightforward log-probability evaluation under PPO. Its main limitation is that it treats all assets symmetrically and does not explicitly encode the hierarchical organization of financial markets.

A *hierarchical* Dirichlet policy instead factorizes allocation into multiple stochastic stages. Let the asset universe be partitioned into G disjoint groups (e.g., sectors), with group g containing N_g assets and $\sum_{g=1}^G N_g = N_t$. At time t, the policy first samples group-level weights

$$v_t \sim \text{Dirichlet}\big(\alpha_t^{(\text{group})}\big), \quad \alpha_t^{(\text{group})} \in \mathbb{R}_{>0}^G, \quad \sum_{g=1}^G v_{t,g} = 1,$$

and then samples conditional intra-group allocations

$$u_{t,g} \sim \text{Dirichlet}\big(\alpha_{t,g}^{(\text{asset})}\big), \quad \alpha_{t,g}^{(\text{asset})} \in \mathbb{R}_{>0}^{N_g}, \quad \sum_{i=1}^{N_g} u_{t,g,i} = 1.$$

Final portfolio weights are obtained by hierarchical composition,

$$w_{t,g,i} = v_{t,g}\,u_{t,g,i}, \qquad \sum_{g=1}^G \sum_{i=1}^{N_g} w_{t,g,i} = 1.$$

This formulation preserves global feasibility while separating sector-level and within-sector allocation decisions, enabling structured exploitation of covariance regularities and improved interpretability via multi-level attribution of risk and capital. However, it introduces additional parameters, nested sampling, and more complex gradient estimation, increasing computational cost and training sensitivity. The flat Dirichlet policy therefore serves as a parsimonious baseline that isolates the effects of Dirichlet action geometry and attention without imposing group structure. We adopt this formulation to clearly attribute performance gains to the parameterization and attention mechanism rather than to hierarchical priors. The framework remains modular, however, and can readily accommodate hierarchical Dirichlet heads, mixture-of-Dirichlet layers, or graph-based allocation modules encoding sectoral or network relationships, which we leave to future work.

6. Experimental design

6.1. Data, universe, and features

Our empirical study uses daily U.S. equity data from the S&P 500 universe, obtained via the Yahoo Finance public API.[1] For each trading day t and ticker n, the panel records adjusted open, high, low, close, and volume (OHLCV), alongside a set of widely used technical indicators. These include 30- and 60-day moving averages, oscillators such as RSI and CCI, directional movement measures $+DI$, $-DI$, and ADX, momentum/trend indicators such as MACD (signal and histogram), and Bollinger bands. The raw files are cleaned by coercing dates into `datetime`, removing invalid rows, and sorting by (Date, ticker). Close-to-close returns

$$r_{t+1,n} = \frac{P_{t+1,n}}{P_{t,n}} - 1$$

serve as price relatives and the basis for reward calculations. Delistings and suspensions are tracked via a tradability mask that prevents allocation to unavailable assets.

Each feature f is pivoted to a Date × ticker matrix and reindexed on the full date–ticker grid. Stacking across features yields a tensor $X \in \mathbb{R}^{T\times N\times F}$ with mask $M \in \{0,1\}^{T\times N}$. To align feature scales while preserving cross-sectional signals, we apply same-day z-scoring within each feature:

$$Z_{t,:,f} = \frac{X_{t,:,f} - \mu_{t,f}}{\sigma_{t,f} + \varepsilon}, \quad \varepsilon = 10^{-8},$$

with NaNs and infinities replaced by zero. This preserves relative momentum information while suppressing level effects.

The sample is split chronologically: letting $t_{\min}$ be the earliest observation,

$$t^\star = t_{\min} + 20 \text{ years},$$

with $\{\text{Date} < t^\star\}$ assigned to training and $\{\text{Date} \geq t^\star\}$ to testing. If the 20-year mark exceeds the sample, the split defaults to the 80th time quantile. This ensures long estimation windows and strict out-of-sample evaluation.

The trading environment is a daily rebalancing simulator (`PortfolioEnv`). The state $S_t \in \mathbb{R}^{W\times N\times F}$ is a rolling window (W=30 days) of standardized features. Actions are portfolio allocations $w_t \in \Delta^{N+1}$ over cash and assets, constrained to the simplex. The mask M ensures weights on unavailable assets are zeroed and renormalized. Turnover costs of $\kappa = 5$ bps per dollar traded are deducted from realized returns. Rewards are log wealth growth net of costs. Cash is always included (`include_cash=True`), enabling dynamic de-risking.

For each trading day we include the prevailing short term risk free rate so that cash earns daily interest consistent with market conditions. We use the three month Treasury bill or the Federal Funds effective rate, convert the annualized series to a daily rate, and align it with our trading calendar. This modification replaces the zero yield assumption and ensures that the evaluation environment captures the relative attractiveness of holding cash versus risky assets in different rate regimes.

6.2. Policy architecture, training protocol, and evaluation

To isolate optimization effects, the policy architecture is fixed across learning rules. Each asset's recent feature history is first summarized by a temporal encoder, then aggregated via a *cross-sectional attention* layer with a global token. This allows the policy to condition on inter-asset dependencies such as sectoral co-movement, factor spillovers, and substitution effects. Unless otherwise noted, hidden width is $d = 64$. The aggregated representation feeds a *flat Dirichlet* head that outputs concentration parameters on the simplex Δ^{N+1} spanning cash and tradable assets; weights are sampled and renormalized after masking.

We compare three reinforcement learning algorithms under this common attention-based policy: Proximal Policy Optimization (PPO) with clipped gradients and GAE; Advantage Actor–Critic (A2C), with MSE-trained critic and no clipping; and REINFORCE, using Monte Carlo returns with a learned baseline. As a benchmark, we evaluate an equal-weight buy-and-hold portfolio of all tradable tickers. Fine-tuning starts from pretrained checkpoints and applies additional on-policy updates; outcomes appear in Section 7.

Training employs Adam with learning rate 3×10^{-4}, gradient clipping at 0.5, and seed 42. Each update processes 128 days with six updates per epoch; minibatches of 256 are split into micro-batches of 32 for PPO/A2C. Discounting and bootstrapping are $\gamma = 0.99$, $\lambda = 0.95$ for PPO/A2C; REINFORCE uses Monte Carlo returns with $\gamma = 0.99$. Policies rebalance daily under $\kappa = 5$ bps turnover cost, running on GPU when available.

Hyperparameter refinement follows a small-budget grid search over learning rates $\{10^{-4}, 5\times 10^{-5}\}$, epochs $\{3, 5\}$, and minibatch sizes $\{128, 256\}$ (micro-batch 32 and updates 6 fixed). Each trial is fine-tuned and evaluated out-of-sample, selecting the best checkpoint per algorithm by Sharpe (alternatives: CAGR or MDD). All runs are seeded and produce reproducible logs and equity curves.

Evaluation is deterministic: stochastic Dirichlet sampling is replaced by its mean. Primary performance is cumulative after-cost wealth

$$W_T = \prod_{t=1}^{T} \left(1 + r_t^{\text{port}}\right),$$

with r_t^{port} the net portfolio return. We report CAGR, annualized return and volatility, Sharpe, Sortino, maximum drawdown (MDD), Calmar, hit rate, average gains/losses, skewness, kurtosis, 5% VaR, 5% CVaR, tail ratio, and terminal wealth.

7. Results

7.1. Baseline performance

Table 2 reports baseline performance, while Fig. 3 plots cumulative wealth over 2020–2025. RL policies consistently outperform buy-and-hold. PPO delivers terminal wealth 2.11 (CAGR 14.6%), followed by A2C (2.06, CAGR 14.0%) and REINFORCE (2.04, CAGR 13.9%). Buy-and-hold achieves only 1.94 (CAGR 12.9%). These incremental differences compound into meaningful long-run gains.

Risk-adjusted returns also favor RL. PPO attains Sharpe 0.73 and Sortino 1.03, compared with buy-and-hold's 0.66 and 0.92. Calmar is likewise higher (0.39 vs. 0.34). The equity curves show that while all methods experienced ≈38% drawdowns during COVID-19 and mid-2022, RL policies recovered faster and maintained a persistent lead. By 2025, PPO compounded nearly 9% higher wealth than buy-and-hold. Importantly, RL strategies remain highly correlated with the benchmark, indicating that gains stem from cross-sectional reallocation rather than aggregate market timing. Nonetheless, persistent large drawdowns highlight vulnerability to systemic shocks. Nonetheless, the persistence of ≈38% drawdowns across all methods highlights that the current variance-style penalty improves efficiency but does not materially alter downside risk or path-dependent drawdowns.

7.2. Fine-tuning results

We next evaluate fine-tuned policies. Each algorithm was re-trained for six additional updates under different optimizer settings. Table 3 reports the best configuration per algorithm, while the analysis compares tuned to baseline.

For PPO, fine-tuning materially improved performance. The optimal configuration, characterized by a learning rate of 5×10^{-5}, 3 epochs, and a minibatch size of 256, achieved a Sharpe ratio of 0.776 and a

[1] Accessed at https://finance.yahoo.com.

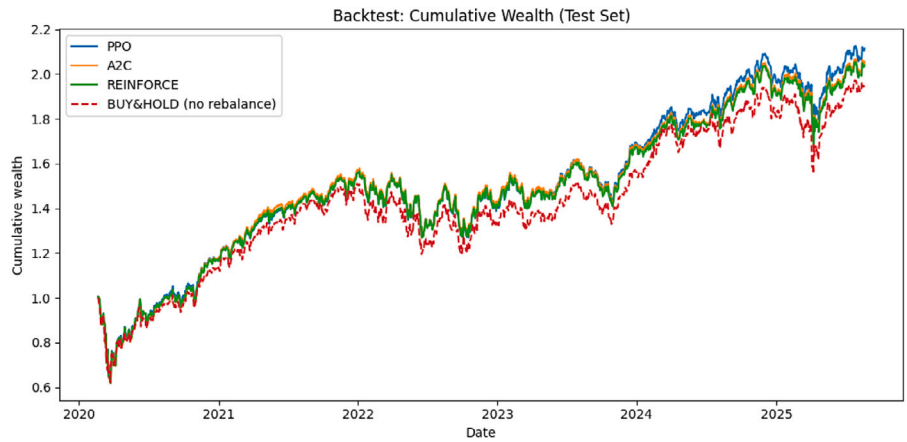

Fig. 3. Backtest cumulative wealth (2020–2025). PPO, A2C, and REINFORCE track aggregate market cycles but sustain higher cumulative returns than buy-and-hold, with faster recovery from drawdowns.

Table 2
Performance comparison of RL policies and buy-and-hold benchmark on the S&P 500 test set.

Algo	Term. Wealth	CAGR	Ann. Ret.	Ann. Vol.	Sharpe	Sortino	Calmar	MDD
PPO	2.1148	0.1462	0.1610	0.2206	0.7298	1.0303	0.3852	−0.3796
A2C	2.0561	0.1404	0.1568	0.2248	0.6975	0.9864	0.3648	−0.3848
REINFORCE	2.0408	0.1388	0.1549	0.2224	0.6963	0.9831	0.3621	−0.3833
Buy&Hold	1.9433	0.1287	0.1460	0.2225	0.6560	0.9207	0.3404	−0.3781

Table 3
Best fine-tuned configuration per algorithm (selected by Sharpe on the test set).

Algo	LR	Epochs	MB	MB_{micro}	ExtraUpd	Sharpe	CAGR/MDD
PPO	5×10^{-5}	3	256	32	6	0.776	15.98%/−37.81%
A2C	5×10^{-5}	3	256	32	6	0.689	13.81%/−38.40%
REINFORCE	5×10^{-5}	3	256	32	6	0.683	13.57%/−38.33%

CAGR of 15.98% with an MDD of −37.8%. This exceeds the baseline PPO Sharpe 0.730 and CAGR 14.6%, indicating that smaller learning rates and larger batches stabilize updates and enhance exploitation of cross-sectional signals. By contrast, A2C and REINFORCE displayed narrow dispersion across the grid: Sharpe clustered around 0.687–0.689 for A2C and 0.675–0.683 for REINFORCE, with CAGRs ≈13.7% and ≈13.5%, respectively. Improvements relative to baseline were marginal. This suggests that simpler policy-gradient methods are less responsive to optimizer tuning and inherently constrained by their variance properties.

Taken together, the fine-tuning study shows that PPO is the most adaptable algorithm in this attention-based portfolio setting, with hyperparameter calibration yielding meaningful gains in Sharpe efficiency and long-run growth. A2C and REINFORCE, while competitive with buy-and-hold, offer limited upside from tuning.

Extended downside risk metrics. Although RL policies enhance long-run returns and Sharpe efficiency, their downside-risk behavior remains broadly similar to the buy-and-hold benchmark. Table 4 reports additional tail-sensitive metrics, including historical 5% VaR, 5% CVaR, tail ratios, and maximum drawdown durations. Across all methods, $CVaR_{5\%}$ lies in a narrow range (−3.25% to −3.33%) and drawdown durations differ only modestly. These results confirm that the variance-style penalty used in our baseline objective primarily improves risk-adjusted performance rather than reducing extreme downside exposure.

Hyperparameter sensitivity and robustness. PPO's heightened sensitivity to hyperparameters stems from its clipped policy-gradient objective, which regulates policy updates through a trust-region–like mechanism controlled by the clipping parameter ϵ. Learning rate, batch size, number of epochs, and clipping threshold jointly determine gradient variance, update magnitude, and convergence stability; small deviations therefore materially affect convergence speed, oscillatory behavior, and sample efficiency. Empirically, this sensitivity manifests as greater dispersion in Sharpe and CAGR outcomes across PPO configurations (Table 3). Lower learning rates and larger minibatches generally improve robustness by reducing gradient noise and mitigating policy collapse, though overly conservative settings can slow adaptation in evolving markets.

Table 4
Extended downside risk metrics.

Algo	MDD	Max DD duration	$VaR_{5\%}$	$CVaR_{5\%}$	Tail ratio	Skew
PPO	−0.3796	373	−0.0184	−0.0325	1.0600	−0.4755
A2C	−0.3848	378	−0.0195	−0.0331	1.0208	−0.4366
REINFORCE	−0.3833	378	−0.0193	−0.0328	1.0131	−0.4590
Buy&Hold	−0.3781	396	−0.0197	−0.0328	1.0032	−0.4626

By contrast, A2C and REINFORCE rely on simpler gradient estimators without clipping and with fewer tunable parameters, yielding smoother but less adaptive optimization trajectories. While they exhibit higher gradient variance and slower asymptotic improvement, their performance is less sensitive to configuration changes. Thus, PPO trades robustness conditional on careful tuning for greater adaptability, whereas A2C and REINFORCE favor stability at the cost of flexibility. Practically, this implies that PPO requires systematic calibration – e.g., rolling-window validation, adaptive early stopping, or dynamic learning-rate schedules – prior to deployment, while A2C and REINFORCE may be preferable in stable or low-frequency settings with limited tuning budget. In nonstationary environments, PPO's tunable surrogate loss offers superior responsiveness provided hyperparameters are selected via structured search or meta-optimization.

Extensions such as population-based training, Bayesian optimization, or gradient-based meta-learning offer promising avenues to further mitigate sensitivity and enhance robustness in live trading systems.

8. Interpretation & economic mechanisms

Our empirical results show that reinforcement learning (RL)–based portfolio policies deliver incremental yet persistent outperformance relative to a naïve buy-and-hold benchmark. Across PPO, A2C, and REINFORCE, the learned policies achieve higher long-run growth and superior Sharpe and Sortino ratios while remaining exposed to aggregate market cycles. These gains do not arise from market timing, but from systematic cross-sectional reallocation: the policies exploit relative strength, momentum, and volatility-adjusted signals to harvest small inefficiencies that compound over time, improving portfolio efficiency without materially altering market exposure.

Downside-risk analysis indicates that the improvement is primarily risk-adjusted rather than tail-protective. Although all RL strategies outperform buy-and-hold in Sharpe and Sortino, their drawdown magnitudes, durations, and tail losses ($\mathrm{VaR}_{5\%}$ and $\mathrm{CVaR}_{5\%}$) remain comparable to the benchmark. This suggests that the current variance-style penalty enhances performance through cross-asset tilts rather than by reducing vulnerability to systemic shocks, motivating extensions with explicit risk-sensitive objectives such as CVaR regularization, volatility targeting, or drawdown-aware reward shaping.

Methodologically, the central contribution is the integration of an *attention-based policy architecture* into the DRL framework. Unlike prior approaches that rely on convolutional or recurrent encoders, our design combines per-asset temporal encoding with cross-sectional attention and a global token, enabling dynamic focus on the most informative assets and features. This structure captures sectoral spillovers, common factor exposures, and relative-value relationships that rigid architectures overlook, better reflecting the interdependent nature of asset returns.

Attention weights provide a transparent measure of cross-asset influence in portfolio construction. Assets or sectors exhibiting stronger co-movement, distinctive volatility, or persistent relative momentum receive greater attention mass, particularly during periods of high return dispersion when a small subset of assets explains most portfolio-level variation. Conversely, more homogeneous conditions yield diffuse attention, corresponding to diversified exposure. In this way, attention captures the evolving network of interdependencies underlying risk and expected return rather than acting as a black box.

The persistence of large drawdowns across all strategies highlights a fundamental limitation: predictive architectures alone cannot resolve the equity risk–return trade-off. Future work should therefore integrate explicit risk constraints alongside attention. Additional extensions include enriching state representations with macroeconomic or alternative data, hybridizing attention-based RL with econometric asset-pricing models, and testing robustness across international markets and longer horizons.

Overall, this study demonstrates that attention mechanisms can be effectively embedded in DRL portfolio policies, yielding both methodological advances and economic insight. By adaptively weighting cross-sectional dependencies, the proposed framework captures richer interactions than prior DRL models and translates them into higher risk-adjusted returns, establishing attention-based architectures as a promising foundation for future research in financial reinforcement learning.

9. Conclusion

This study proposes a deep reinforcement learning framework for dynamic portfolio optimization that combines a Dirichlet policy with cross-sectional attention. The design enforces feasibility on the simplex, incorporates tradability constraints and transaction costs, and links learning objectives to classical mean–variance trade-offs. Attention enables adaptive weighting of sectoral co-movements and relative-value signals, improving information extraction from high-dimensional equity data.

Empirical results on two decades of S&P 500 data show that attention-enhanced Dirichlet policies outperform equal-weight and reinforcement learning baselines in terminal wealth and risk-adjusted returns without relying on market timing. The gains arise from systematic cross-sectional reallocations exploiting relative strength, momentum, and volatility-adjusted inefficiencies.

While the framework improves efficiency and interpretability, persistent drawdowns indicate that predictive architectures alone cannot resolve the equity risk–return trade-off. Future work should incorporate explicit risk-sensitive objectives, richer macroeconomic state information, and broader market validation, as well as explore hybrid approaches combining reinforcement learning with econometric models.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The data used in this study are publicly available. Financial market data were obtained from **Yahoo Finance**. Specifically, daily transaction and pricing data for the **500 constituent stocks of the S&P 500** were accessed through Yahoo Finance's open API and data portal, available at:https://finance.yahoo.com/.

References

Almgren, R., & Chriss, N. (2000). Optimal execution of portfolio transactions. *Journal of Risk*, *3*, 5–39.

Bailey, D. H., Borwein, J. M., Lopez de Prado, M., & Zhu, Q. J. (2017). The probability of backtest overfitting. *Journal of Computational Finance*, *20*(4), 39–69.

Bailey, D. H., Lopez de Prado, M., Molino, J., & Dash, J. (2016). Deflated sharpe ratio: Correcting for selection bias and multiple testing. *Journal of Portfolio Management*, *42*(2), 59–70.

Bollerslev, T. (1986). Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, *31*(3), 307–327.

Deng, Y., Bao, F., Kong, Y., Ren, Z., & Dai, Q. (2016). Deep direct reinforcement learning for financial signal representation and trading. *IEEE Transactions on Neural Networks and Learning Systems*, *28*(3), 653–664.

Fama, E. F., & French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, *33*(1), 3–56.

Gatheral, J. (2010). No-dynamic-arbitrage and market impact. *Quantitative Finance*, *10*(7), 749–759.

Grossman, S. J., & Zhou, Z. (1993). Optimal investment strategies for controlling drawdowns. *Mathematical Finance*, *3*(3), 241–276.

Harvey, C. R., Liu, Y., & Zhu, H. (2016). ... and the cross-section of expected returns. *Review of Financial Studies*, *29*(1), 5–68.

Jegadeesh, N., & Titman, S. (1993). Returns to buying winners and selling losers: Implications for stock market efficiency. *Journal of Finance*, *48*(1), 65–91.

Jiang, Z., Xu, D., & Liang, J. (2017). Deep reinforcement learning for quantitative trading. In *Proceedings of the 24th ACM SIGKDD international conference on knowledge discovery and data mining* (pp. 575–583).

Kelly, J., Jr. (1956). A new interpretation of information rate. *Bell System Technical Journal*, *35*(4), 917–926.

Ledoit, O., & Wolf, M. (2004). Honey, I shrunk the sample covariance matrix. *The Journal of Portfolio Management*, *30*(4), 110–119.

Lim, B., Arik, S. Ö., Loeff, N., & Pfister, T. (2021). Temporal fusion transformers for interpretable multi-horizon time series forecasting. *International Journal of Forecasting*, *37*(4), 1748–1764.

Markowitz, H. (1952). Portfolio selection. *The Journal of Finance*, *7*(1), 77–91.

Merton, R. C. (1971). Optimum consumption and portfolio rules in a continuous-time model. *Journal of Economic Theory*, *3*(4), 373–413.

Moody, J., & Saffell, M. (2001). Learning to trade via direct reinforcement. *Vol. 12*, In *IEEE transactions on neural networks* (pp. 875–889).

Nevmyvaka, Y., Feng, Y., & Kearns, M. (2006). Reinforcement learning for optimized trade execution. In *Proceedings of the 23rd International Conference on Machine Learning* (pp. 673–680).

Rockafellar, R. T., & Uryasev, S. (2000). Optimization of conditional value-at-risk. *Journal of Risk*, *2*(3), 21–42.

Romano, J. P., & Wolf, M. (2005). Stepwise multiple testing as formalized data snooping. *Econometrica*, *73*(4), 1237–1282.

Schulman, J., Wolski, F., Dhariwal, P., Radford, A., & Klimov, O. (2017). Proximal policy optimization algorithms. arXiv preprint arXiv:1707.06347.

Silva, Y., & Almeida Filho, A. (2023). A systematic literature review on solution approaches for the index tracking problem in the last decade. *Expert Systems with Applications*, *216*, Article 119436.

White, H. (2000). A reality check for data snooping. *Econometrica*, *68*(5), 1097–1126.

Yang, H., Liu, X.-Y., Zhong, S., Walid, A., Yu, R., & Li, B. (2020). Deep reinforcement learning for automated stock trading: An ensemble strategy. *Quantitative Finance*, *20*(9), 1419–1444.

Yuen, M.-C., Ng, S.-C., & Leung, M.-F. (2021). Metaheuristics for sparse index-tracking problem: A case study on FTSE 100. *Journal of Physics: Conference Series*, *1828*(1), Article 012111.

Yuen, M.-C., Ng, S.-C., Leung, M.-F., & Che, H. (2022). A metaheuristic-based framework for index tracking with practical constraints. *Complex and Intelligent Systems*, *8*(6), 4571–4586.

Zhang, X.-Y., et al. (2020). FinRL: Deep reinforcement learning framework to automate trading in quantitative finance. arXiv preprint arXiv:2011.09607.

Zhou, H., Zhang, S., Peng, J., Zhang, S., Li, J., Xiong, H., & Zhang, W. (2021). Informer: Beyond efficient transformer for long sequence time-series forecasting. *Vol. 35*, In *Proceedings of the AAAI conference on artificial intelligence* (pp. 11106–11115).